Data Scientist Career Roadmap (India, 0-8 years)

A comprehensive guide to building your data science career in India, from fresher to senior level, with salary insights, learning paths, and portfolio development strategies.

Your Journey: Fresher to 8 Years

1. **Phase 0: Solid Foundations (0-6 months)**
   Build core skills in Python, SQL, statistics, and visualization tools

2. **Phase 1: Core ML for DS (6-12 months)**
   Master machine learning fundamentals and feature engineering

3. **Phase 2: Experimentation (1-2 years)**
   Develop expertise in A/B testing and product analytics

4. **Phase 3: Specialization (2-3 years)**
   Choose a focus area and learn production deployment

5. **Phase 4: Senior DS (3-5 years)**
   Own end-to-end problems and mentor juniors

6. **Phase 5: Staff/Lead DS (5-8 years)**
   Drive strategy and organizational influence

Each phase builds on the previous one, creating a progressive learning journey with clear deliverables and skill milestones.

Phase 0: Solid Foundations (0-6 months)

Core Technical Skills

- Python (NumPy, pandas)
- SQL (joins, windows)
- Git version control
- Linux basics

Statistics Fundamentals

- Probability concepts
- Statistical distributions
- Hypothesis testing
- Confidence intervals

Data Visualization

- Matplotlib/Plotly
- Seaborn
- Power BI/Tableau

Expected Outputs

- 2 mini-analyses
- 1 dashboard
- ATS-optimized resume

Project ideas: Sales/Leads dashboard, A/B test simulation, customer churn exploratory data analysis

Phase 1: Core ML for DS (6–12 months)

Machine Learning Algorithms

- Linear/logistic regression
- Decision trees
- Random forest
- Boosting (XGBoost/LightGBM)
- Regularization techniques

Feature Engineering

- Leakage checks
- Missing data handling
- Outlier treatment
- Scaling/encoding methods

Model Validation

- Cross-validation strategies
- Metrics (ROC-AUC, F1, MAE/RMSE)
- Model calibration

Expected Outputs: 2 end-to-end notebooks + 1 FastAPI inference endpoint

Project Ideas: Demand forecasting, customer segmentation, lead scoring model

Phase 2: Experimentation & Product Analytics (1-2 years)

Causal Inference Basics

- A/B test design principles
- CUPED (Controlled-experiment Using Pre-Experiment Data)
- Introduction to difference-in-differences

Experiment Platforms

- Metrics definition and tracking
- Guardrails implementation
- Sample size calculation
- Statistical power analysis
- Lift analysis techniques

Analytics Engineering

- dbt basics
- Data quality checks
- Documentation standards

Expected Outputs

Experiment analysis case study + metrics specification document

Project Ideas: Pricing A/B test, recommender system impact analysis, notification optimization experiments

Phase 3: Specialize & Ship to Production (2-3 years)

Choose Your Specialization

- Forecasting
- Recommendation systems
- NLP analytics
- Risk/fraud detection
- Marketing science

MLOps Fundamentals

- Model packaging
- MLflow/Weights & Biases
- Feature stores (introduction)
- Batch vs streaming pipelines

Stakeholder Skills

- Project scoping
- Product requirement docs
- Data storytelling
- Decision-first dashboards

Expected Outputs: Productionized model + business impact study

At this stage, you should be able to take a data science solution from concept to production, with clear documentation of business value.

Phase 4: Senior Data Scientist (3-5 years)

Technical Leadership

- Own ambiguous problems end-to-end
- Develop experiment roadmaps
- Implement causal ML at scale
- Build time-series solutions at scale

Model Governance

- Monitoring systems
- Drift detection
- Documentation standards

Team Contributions

- Mentor junior data scientists
- Lead code and model reviews
- Partner effectively with Product, Engineering, and Design teams

Expected Outputs

Multiple shipped projects with measurable business impact

As a Senior DS, you'll balance technical depth with broader organizational impact, becoming a go-to resource for complex data problems.

Phase 5: Staff/Lead DS (5-8 years)

Strategic Leadership

- Analytics strategy development
- Experimentation roadmaps
- KPI trees and frameworks
- Data contracts establishment

Organizational Influence

- Cross-functional leadership
- Vendor/tooling selection
- Hiring and team building
- Mentoring program development

Key Deliverables

- Technical playbooks
- Request for Comments (RFCs)
- Quarterly strategy documents
- Business outcome reports

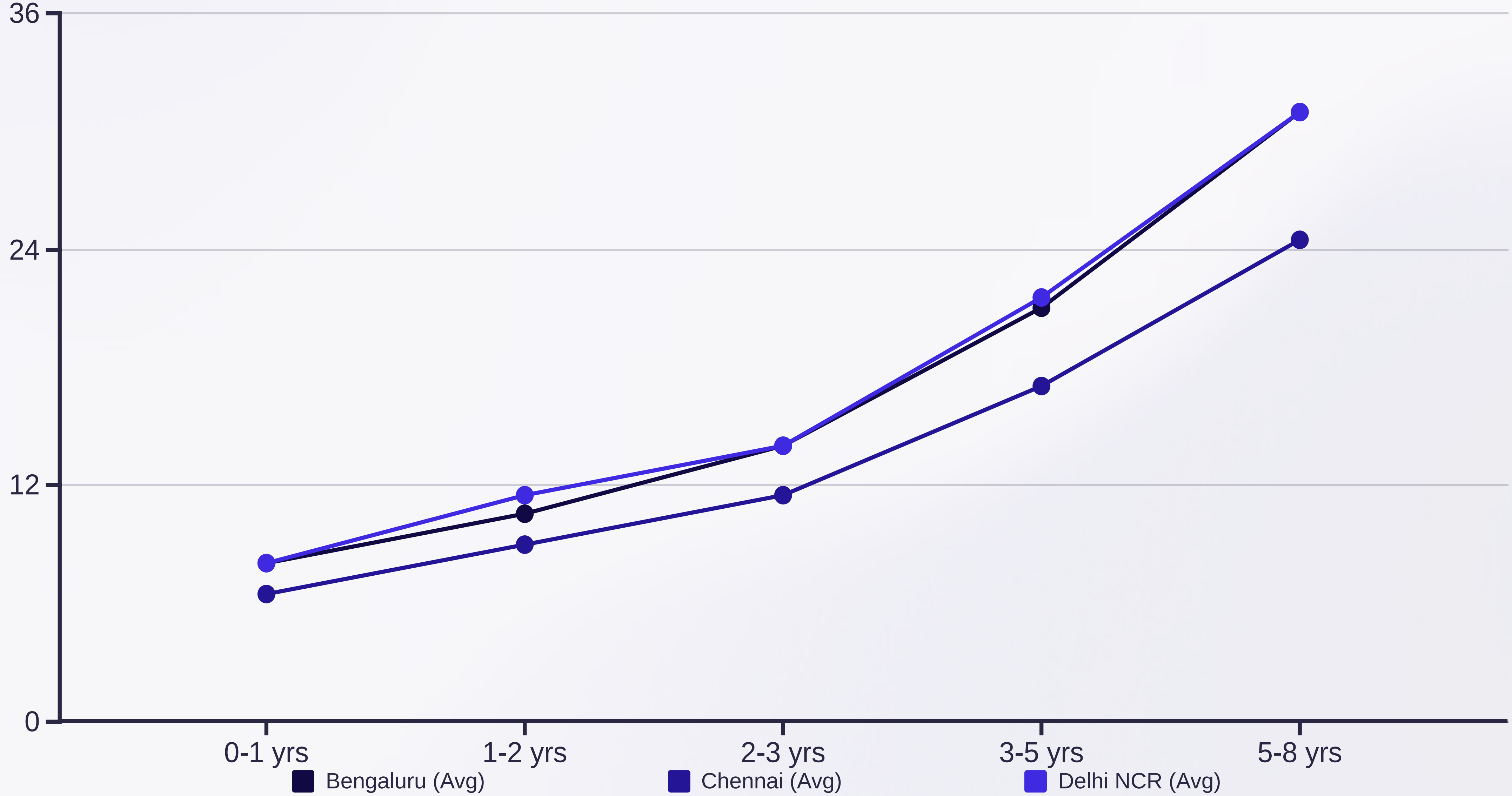
At this level, you'll shape how the entire organization approaches data science, balancing technical excellence with business strategy and team development.

City-wise India Salaries (LPA)

Salary ranges are indicative; strong SQL/stats/experimentation skills and production experience push you to the top of the band.

City	0-1 yrs	1-2 yrs	2-3 yrs	3-5 yrs	5-8 yrs
Bengaluru	6-10 L	8-13 L	11-17 L	16-26 L	24-38 L
Delhi NCR	6-10 L	9-14 L	11-17 L	16-27 L	24-38 L
Hyderabad	5-9 L	8-13 L	10-16 L	15-24 L	22-34 L
Mumbai	5-9 L	8-12 L	10-15 L	15-23 L	22-33 L
Pune	5-9 L	8-12 L	10-15 L	14-22 L	20-32 L
Chennai	5-8 L	7-11 L	9-14 L	13-21 L	19-30 L

Salary Growth Trajectory



The chart shows average salary progression (in LPA) across three major tech hubs in India. Note the significant jump between the 2-3 year and 3-5 year experience bands, highlighting the value of reaching senior data scientist level.

Essential Skills to Develop

Math & Statistics

- Probability theory
- Statistical distributions
- Central Limit Theorem
- Hypothesis testing
- ANOVA
- Regression diagnostics
- Time-series analysis

SQL & Data Engineering

- Complex joins
- Window functions
- Subqueries & CTEs
- Data modeling
- Query performance
- Data quality checks

Python & ML

- Pandas/NumPy/Scikit-learn
- Testing frameworks
- Package development
- Feature engineering
- Model selection
- Calibration techniques
- Interpretability (SHAP/PDP)

These core technical skills form the foundation of your data science career. Continuously deepen your expertise in these areas as you progress.

More Essential Skills

Experimentation

- A/B test design
- Metrics definition
- CUPED methodology
- Statistical power analysis
- Guardrail implementation

Visualization

- Data storytelling
- Dashboard development
- Tableau/Power BI
- Streamlit applications

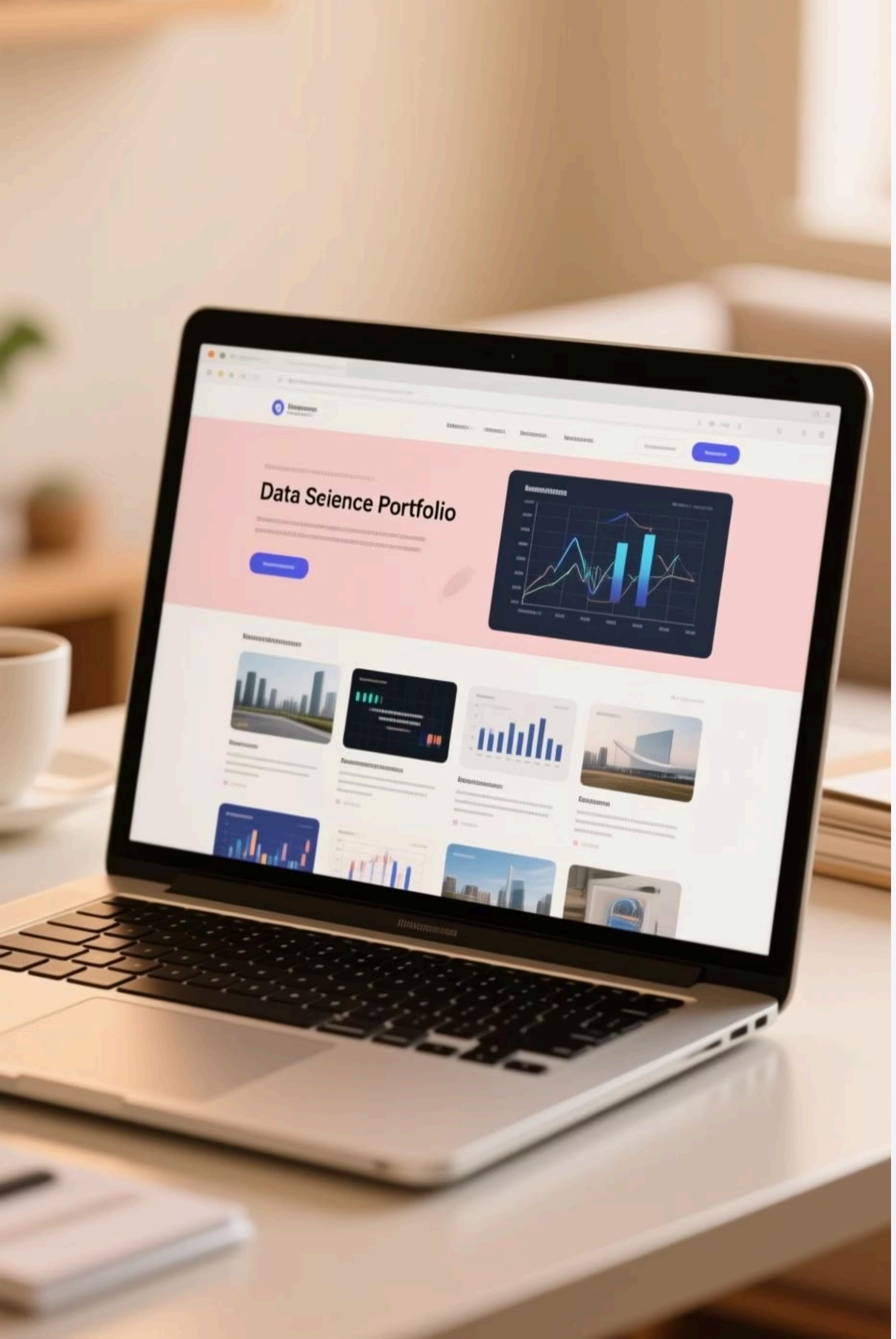
MLOps Lite

- FastAPI development
- MLflow/Weights & Biases
- Batch scoring pipelines
- Airflow/Prefect workflows

Domain Knowledge

- Marketing/growth
- Product funnels
- Risk/fraud detection
- Supply chain
- Finance

Beyond technical skills, domain expertise and production capabilities will set you apart as you advance in your career.



Project Ideas for Your Portfolio

Analytics Projects

- Cohort & retention analysis dashboard
- Revenue anomaly detection system
- Customer segmentation with actionable insights

Forecasting Projects

- Multi-store demand prediction
- Manpower planning optimization
- Lead/admissions forecasting

Recommendation Systems

- Content-based recommender
- Collaborative filtering implementation
- Offline/online evaluation framework

Advanced Applications

- Causal ML for campaign uplift modeling
- Self-serve A/B testing dashboard
- KPI guardrails monitoring system
- Churn prediction playbook

Career Pivot Options (After ~3 Years)

Senior DS → Staff DS/Manager

Focus on experimentation strategy, KPI development, and multi-quarter initiatives. Lead teams and develop organizational data science capabilities.

DS → ML Engineer

Specialize in serving pipelines, CI/CD, feature stores, and latency/cost optimization for machine learning systems.

DS → GenAI-focused DS

Work with embeddings/retrieval, evaluation frameworks, summarization techniques, and insights co-pilots using generative AI.

DS → Product Analytics Lead

Build experimentation platforms, establish metric governance, and drive decision science across product teams.

After gaining solid experience as a data scientist, you can pivot to specialized roles that align with your interests and strengths.

12-Week Upskill Track

1 Weeks 1-3: Data Engineering Focus

- Advanced SQL and performance optimization
- dbt implementation for transformation
- Data quality frameworks
- Documentation best practices

2 Weeks 4-6: Experimentation at Scale

- CUPED methodology for variance reduction
- Sequential testing frameworks
- Comprehensive case study development

3 Weeks 7-9: Advanced Modeling

- Deep dive into forecasting techniques
- Recommendation system architectures
- Reproducible pipelines with MLflow

4 Weeks 10-12: Productionization

- FastAPI for batch/real-time inference
- Monitoring and drift detection
- Incident response and postmortems

This structured learning plan helps experienced data scientists level up their skills in a systematic way.

Weekly Study Plan (First 6 Weeks)

Week	Focus	Goals	Output
1	Python/SQL refresh + Git	Data handling & window functions	Mini ETL + SQL exercises
2	Stats & EDA	Hypothesis tests & viz best practices	EDA notebook on public dataset
3	Regression & classification	Baselines + metrics	Model comparison report
4	Feature engineering	Lift via features; avoid leakage	Feature importance + checklist
5	Boosting (XGB/LGBM)	Tuning; CV; calibration	Tuned model + calibration plot
6	Experimentation intro	A/B design; power; CUPED	Experiment plan & analysis

This structured weekly plan provides clear focus areas, goals, and tangible outputs to track your progress during the first half of your 12-week learning journey.

Weekly Study Plan (Weeks 7–12)

Week	Focus	Goals	Output
7	Time-series forecasting	Classical models & exogenous vars	Forecast with backtests
8	Recommender systems	Similarity & CF; offline eval	Top-N recommender demo
9	Productionization	FastAPI + batch scoring	Deployed endpoint or batch job
10	Monitoring & drift	Evidently/WhyLabs; alerts	Monitoring dashboard
11	Dashboards & storytelling	Decision-first viz	Power BI/Tableau/Streamlit app
12	Capstone & case study	Business impact narrative	Case study + README + Loom

The second half of your 12-week plan focuses on advanced modeling, production deployment, and creating polished portfolio pieces that demonstrate your end-to-end capabilities.

Portfolio Checklist

1 4-6 Diverse Projects

Include at least one project from each category:

- Exploratory data analysis
- Forecasting implementation
- Recommendation system
- Experiment analysis

Ensure at least one project is deployed and one includes a dashboard.

2 Professional Documentation

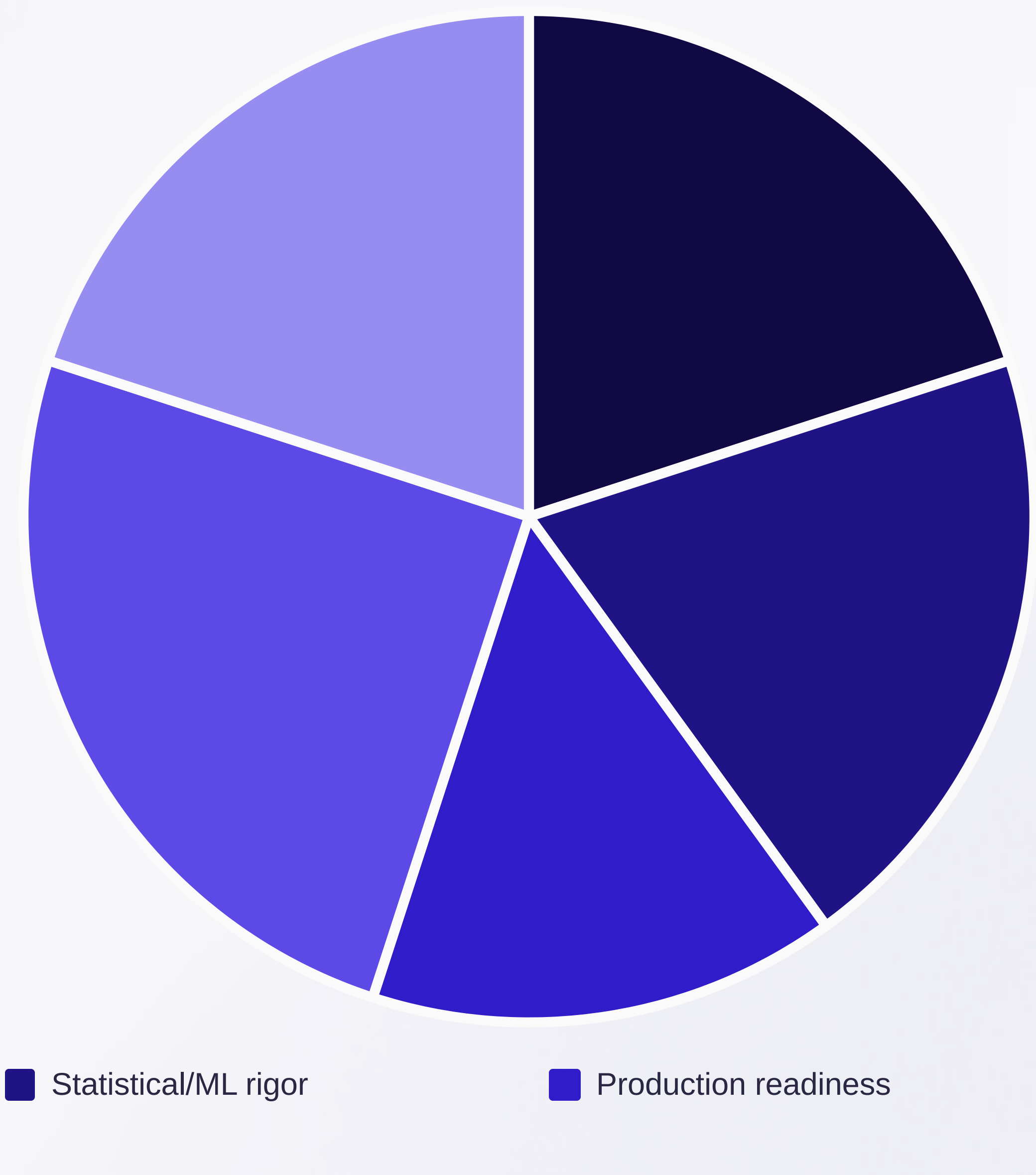
Each project should have a README covering:

- Problem statement and business context
- Data description and sources
- Methodology and approach
- Evaluation metrics and results
- Key decisions and trade-offs
- Future improvement opportunities

3 Career Materials

- ATS-optimized resume
- LinkedIn profile with featured projects
- Case studies highlighting business impact
- Ethics considerations where applicable

Portfolio Evaluation Rubric



Use this 100-point rubric to evaluate your portfolio before applying to jobs. Note that communication and storytelling (25 points) is weighted slightly higher than technical aspects, highlighting the importance of explaining your work clearly to non-technical stakeholders.

Strong portfolios demonstrate not just technical skills but also business acumen and the ability to drive decisions through data.

Your Data Science Journey Starts Now

Build Foundations

Master the core technical skills in Python, SQL, and statistics that form the foundation of data science work.

Create Portfolio

Develop projects that demonstrate your skills and showcase your ability to deliver business value through data.

Gain Experience

Progress through the career phases, continuously learning and specializing in areas that interest you.

Lead & Influence

As you advance, develop leadership skills to drive data strategy and mentor the next generation.

This roadmap provides a structured path from fresher to senior data scientist in India. Remember that consistent learning and practical application are key to success in this rapidly evolving field.